Software Requirements Specification (SRS)

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Arfath Event Management System (EMS)

Version 1.0

Document Control

Version	Date	Author	Description
1.0	2026-03-13	Arfath EMS Team	Initial SRS – covers MVP and full system requirements

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete set of requirements for the Arfath Event Management System (EMS). It serves as a communication medium between stakeholders, developers, and testers, and provides the basis for design, implementation, and validation. The document covers both the Minimum Viable Product (MVP) to be delivered in two weeks and the full system planned over subsequent iterations.

1.2 Scope

Arfath EMS is a web-based platform that enables event organizers to create and manage events, sell tickets, engage attendees, and analyze performance. Attendees can discover events, register, pay, and participate both onsite and virtually. The system is built using Python (Flask), Firebase Firestore, Firebase Authentication, PayPal, and Docker, and is designed for deployment on a local server.

The MVP focuses on core functionality: event creation, ticket sales with PayPal, email confirmation with QR codes, and a basic organizer dashboard. Subsequent releases add agenda management, attendee portals, onsite check-in, promotional tools, CRM integration, networking, and advanced analytics.

1.3 Definitions and Acronyms

Term	Definition
EMS	Event Management System
MVP	Minimum Viable Product
Organizer	User who creates and manages events
Attendee	User who registers and participates in events
Admin	System administrator with full privileges
Speaker	Individual presenting at a session
Sponsor	Organization financially supporting an event
Exhibitor	Company with a booth at an event
QR Code	Quick Response code used for ticket validation and check-in
CRM	Customer Relationship Management (e.g., Salesforce, HubSpot)
Firestore	NoSQL document database provided by Firebase
Firebase Auth	Authentication service by Firebase
Flask	Python web framework used for the backend
PayPal	Payment gateway for processing transactions
Docker	Containerization platform for deployment

1.4 References

- Arfath EMS Feature List (previous document)
- Arfath EMS ER Diagram (Firestore logical model)
- Arfath EMS Sprint Roadmap (2-week increments)
- Firebase Documentation
- PayPal Developer Documentation

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2. Overall Description

2.1 Product Perspective

Arfath EMS is a new, standalone system. It replaces manual event management processes with an integrated digital platform. The system interacts with external services:

- **Firebase Authentication** – for user identity.
- **Firebase Firestore** – for data persistence.

- **PayPal** – for payment processing.
- **SendGrid / SMTP** – for email notifications.
- **YouTube / Zoom** – for embedding live streams (future).
- **HubSpot / Salesforce** – for CRM synchronization (future).

The system is deployed using Docker containers on a local server, accessible via a local network.

2.2 User Characteristics

Stakeholder	Description	Key Responsibilities / Needs
Organizer	Event planners who create and manage events.	Create events, configure tickets, view registrations, manage agenda, check in attendees, run reports.
Attendee	End-users who register and attend events.	Discover events, register, pay, access agenda, network, provide feedback.
Admin	Platform-level superuser.	Manage users, oversee system settings, handle support issues.
Speaker	Individuals presenting at sessions.	Provide profile, upload materials, engage with attendees (future).
Sponsor	Organizations sponsoring events.	Receive recognition, access leads (future).
Exhibitor	Companies with booths.	Manage booth, capture attendee leads (future).

2.3 Assumptions and Dependencies

- Users have internet access and a modern web browser.
- Organizers will provide accurate event details (dates, venue, sessions).
- Firebase services (Auth, Firestore) are available and reliable.
- PayPal merchant account is configured and active.
- Email service (SendGrid or SMTP) is properly set up.
- The local server running Docker has sufficient resources (CPU, RAM, disk).
- Docker and Docker Compose are installed on the deployment machine.

2.4 Use Case Overview

The following diagram illustrates the high-level use cases for the system. (A textual representation is provided below; a graphical diagram can be added in the final document.)

Actors: Organizer, Attendee, Admin, PayPal (external), Email Service (external).

Key Use Cases:

- UC1: Register / Login
- UC2: Create Event (Organizer)
- UC3: Manage Ticket Types (Organizer)
- UC4: Browse Events (Attendee)
- UC5: Register for Event (Attendee)
- UC6: Process Payment (PayPal)
- UC7: View Registrations (Organizer)
- UC8: Check In Attendee (Organizer)
- UC9: Manage Agenda (Organizer)
- UC10: Save Sessions (Attendee)
- UC11: Generate Reports (Organizer)
- UC12: Sync with CRM (Organizer/Admin)

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3. System Features and Requirements

Requirements are tagged with a unique ID and a priority (**Must**, **Should**, **Could**). The **MVP** column indicates whether the feature is included in the initial two-week release.

3.1 User Management & Authentication

ID	Requirement	Priority	MVP
REQ-101	Users shall be able to sign up using email and password (via Firebase Auth).	Must	Yes
REQ-102	Users shall be able to log in and log out securely.	Must	Yes
REQ-103	Users shall have a role (organizer, attendee, admin) assigned at signup (default attendee).	Must	Yes
REQ-104	Users shall be able to reset their password via email.	Must	Yes
REQ-105	Users shall be able to update their profile (name, phone, profile picture).	Should	No
REQ-106	Organizers can invite team members with role-based permissions (view-only, editor).	Could	No
REQ-107	Admins can view and manage all users (suspend, change roles).	Should	No

3.2 Event Management

ID	Requirement	Priority	MVP
REQ-201	Organizers shall be able to create an event with name, description, start/end dates, and venue.	Must	Yes
REQ-202	Organizers shall be able to edit and delete draft events.	Must	Yes
REQ-203	Organizers shall be able to publish an event to make it publicly visible.	Must	Yes
REQ-204	Organizers shall be able to set an event as cancelled or completed.	Should	No
REQ-205	Organizers shall be able to upload a cover image and set branding colors.	Should	No
REQ-206	Organizers shall be able to clone an existing event to use as a template.	Could	No

3.3 Venue Management

ID	Requirement	Priority	MVP
REQ-301	Organizers shall be able to add, edit, and delete venues (name, address, capacity, contact info).	Must	Yes
REQ-302	Venues can optionally have a floor plan image (upload).	Should	No

3.4 Ticketing & Registration

ID	Requirement	Priority	MVP
REQ-401	Organizers shall be able to create multiple ticket types per event (name, price, quantity, max per order).	Must	Yes
REQ-402	Ticket types can have sales start/end dates.	Should	No
REQ-403	Organizers shall be able to create promo codes (percentage/fixed discount, validity period, usage limits).	Should	No
REQ-404	Organizers shall be able to define custom registration fields (text, dropdown, checkbox) with conditional logic.	Should	No
REQ-405	Public users shall be able to browse published events and view details.	Must	Yes
REQ-406	Attendees shall be able to select ticket type, quantity, and apply promo code during registration.	Must	Yes
REQ-407	System shall validate ticket availability and promo code eligibility.	Must	Yes

ID	Requirement	Priority	MVP
REQ-408	On successful payment, system shall create a confirmed registration and generate a unique QR code.	Must	Yes
REQ-409	Attendees shall receive a confirmation email with QR code (as image or link).	Must	Yes
REQ-410	Attendees shall be able to view their tickets (with QR code) in their profile.	Should	No
REQ-411	Attendees shall be able to transfer tickets to another person (if allowed by organizer).	Could	No
REQ-412	System shall support waitlisting when tickets are sold out.	Could	No
REQ-413	Organizers shall be able to process refunds (partial/full) from the dashboard.	Could	No

3.5 Payment Processing

ID	Requirement	Priority	MVP
REQ-501	System shall integrate with PayPal (PayPal Checkout) to process payments.	Must	Yes
REQ-502	On payment completion, PayPal shall send an IPN/webhook to confirm the transaction.	Must	Yes
REQ-503	System shall update registration status based on webhook confirmation.	Must	Yes
REQ-504	Payment records (amount, transaction ID, status) shall be stored in Firestore.	Must	Yes
REQ-505	System shall support multiple currencies (future).	Could	No
REQ-506	System shall handle payment failures gracefully and allow retry.	Should	No

3.6 Attendee Portal

ID	Requirement	Priority	MVP
REQ-601	Attendees shall have a dashboard showing their upcoming and past events.	Must	No (MVP+1)
REQ-602	Attendees shall be able to view event agenda and mark sessions as “saved”.	Should	No
REQ-603	Attendees shall be able to access session details (speakers, description, streaming link).	Should	No

ID	Requirement	Priority	MVP
REQ-604	Attendees shall be able to update their profile information.	Should	No

3.7 Agenda & Session Management

ID	Requirement	Priority	MVP
REQ-701	Organizers shall be able to create, edit, and delete sessions (title, description, start/end, track, location).	Should	No
REQ-702	Organizers shall be able to manage speakers (add, edit, bio, photo) and assign them to sessions.	Should	No
REQ-703	Organizers shall be able to provide a streaming URL for virtual sessions.	Should	No
REQ-704	Public event page shall display the agenda with session details.	Should	No
REQ-705	Attendees shall be able to view live stream from session page (embed or link).	Should	No

3.8 Onsite Check-In

ID	Requirement	Priority	MVP
REQ-801	Organizers shall have a web-based check-in page with QR code scanner (using device camera).	Should	No
REQ-802	Scanning a valid QR code shall mark the attendee as checked in (timestamp, staff ID).	Should	No
REQ-803	Check-in status shall be visible in the organizer dashboard.	Should	No
REQ-804	Organizers shall be able to manually check in attendees (search by name/email).	Should	No
REQ-805	System shall support offline check-in with sync later (future).	Could	No

3.9 Marketing & Communication

ID	Requirement	Priority	MVP
REQ-901	Organizers shall be able to send email campaigns to segmented attendee lists (by ticket type, check-in status).	Could	No

ID	Requirement	Priority	MVP
REQ-902	System shall send automated emails: registration confirmation, reminders, post-event thanks.	Must (confirmation)	Yes (confirmation)
REQ-903	Attendees shall be able to manage their notification preferences.	Should	No
REQ-904	System shall support SMS notifications (future).	Could	No

3.10 Analytics & Reporting

ID	Requirement	Priority	MVP
REQ-1001	Organizers shall see a dashboard with key metrics: total registrations, revenue, check-ins.	Must (basic)	Yes (counts)
REQ-1002	Organizers shall be able to view a list of registrations and export to CSV.	Must	Yes
REQ-1003	Organizers shall be able to view charts (registrations over time, sales by ticket type).	Should	No
REQ-1004	Organizers shall be able to generate custom reports (PDF/Excel) on attendance, demographics, session popularity.	Should	No
REQ-1005	System shall provide predictive insights (popular sessions, attrition risk) – future.	Could	No

3.11 Integrations

ID	Requirement	Priority	MVP
REQ-1101	System shall sync attendee data with CRM (HubSpot/Salesforce) via API (organizer configurable).	Could	No
REQ-1102	System shall provide webhooks for external systems to subscribe to events (new registration, check-in).	Could	No
REQ-1103	System shall allow embedding of YouTube/Zoom live streams.	Should	No

3.12 Post-Event Features

ID	Requirement	Priority	MVP
REQ-1201	Organizers shall be able to send automated thank-you emails with feedback survey link.	Should	No
REQ-1202	System shall generate attendance certificates (custom template) and email to attendees.	Could	No
REQ-1203	Event data shall be archived and available for historical analysis.	Should	No

3.13 Admin Functions

ID	Requirement	Priority	MVP
REQ-1301	Admins shall be able to view and manage all users (suspend, change roles).	Should	No
REQ-1302	Admins shall have access to system logs and audit trails.	Could	No
REQ-1303	Admins shall be able to configure global settings (payment gateways, email providers).	Should	No

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4. Non-Functional Requirements

4.1 Performance

ID	Requirement	Target
NFR-01	Page load time for public event list and registration page shall be < 3 seconds (under normal load).	
NFR-02	Registration and payment flow shall complete within 5 seconds (excluding PayPal redirect).	
NFR-03	The system shall support at least 50 concurrent users during MVP, scaling to 500+ in later releases.	
NFR-04	Firestore queries shall use appropriate indexes to ensure response times < 200ms for common queries.	

4.2 Security

ID	Requirement
NFR-05	All communication shall be over HTTPS (SSL/TLS).
NFR-06	Passwords shall be securely hashed via Firebase Auth (never stored in Firestore).
NFR-07	Payment information shall be handled directly by PayPal; no card data stored in our system.
NFR-08	Firebase Firestore security rules shall restrict access: organizers can only modify their own events; attendees can only read their own registrations.
NFR-09	Flask routes shall validate user authentication and authorization (role-based access).
NFR-10	Environment variables (API keys, secrets) shall not be hard-coded; use `.env` and Docker secrets.

4.3 Usability

ID	Requirement
NFR-11	The user interface shall be responsive and work on desktop, tablet, and mobile browsers.
NFR-12	Forms shall provide clear validation messages and error handling.
NFR-13	The organizer dashboard shall present data in an intuitive, visual manner (charts, summaries).
NFR-14	The system shall support keyboard navigation and screen readers for basic accessibility.

4.4 Reliability

ID	Requirement
NFR-15	System uptime target: 99.5% (excluding planned maintenance).
NFR-16	Critical errors (payment failures, registration issues) shall be logged and alerted.
NFR-17	Firestore and Firebase Auth are managed services with high availability; application shall handle temporary outages gracefully (e.g., retry logic).

4.5 Scalability

ID	Requirement
NFR-18	The application shall be stateless (session data stored in Firebase or client-side cookies) to allow horizontal scaling.
NFR-19	Firestore automatically scales; design shall avoid hotspots (e.g., using distributed counters for registration counts).

4.6 Maintainability

ID	Requirement
NFR-20	Code shall follow PEP 8 standards and be well-documented.
NFR-21	The project shall use Git for version control with clear commit messages.
NFR-22	Configuration shall be externalized (environment variables, Docker Compose).
NFR-23	Unit tests shall cover critical business logic (registration, promo codes, check-in).

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5. Constraints and Assumptions

5.1 Constraints

- Development must use Python (Flask), Firebase (Auth, Firestore), PayPal, and Docker.
- The MVP must be delivered within two weeks.
- The system is deployed on a local server using Docker; no cloud hosting initially.
- All data must reside in Firestore (NoSQL); relational joins are not available.

5.2 Assumptions

- Organizers have basic technical proficiency to manage events.
- Attendees have access to email and a modern browser.
- PayPal merchant account is active and configured to send IPNs to the server.
- The local server has a static IP or hostname accessible on the local network.
- Docker and Docker Compose are installed and properly configured.

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6. Future Scope

The following features are planned for releases beyond the initial 8-week cycle:

- Mobile application (React Native) with offline capabilities.
- AI-powered session recommendations and chatbots.
- Sponsors and exhibitors management with lead capture (QR scanning at booths).
- Multi-currency and multi-language support.
- White-labeling: custom domains and full branding control.
- Advanced networking: matchmaking, virtual meeting rooms.
- Sustainability tracking (carbon footprint estimation).
- Accessibility enhancements (WCAG 2.1 AA compliance).
- Blockchain-based ticketing for anti-counterfeiting.

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7. Appendices

(Optional – can include use case diagrams, wireframe references, or detailed data dictionary.)